

# Hermenegildo Jerónimo dos Santos

## Full-Stack AI Engineer · Real-Time & Human-in-the-Loop Systems

[hermeny7@hotmail.com](mailto:hermeny7@hotmail.com) · Covilhã, Portugal · [hermenegildosantos.com](https://hermenegildosantos.com) · [linkedin.com/in/hermenegildosantos](https://linkedin.com/in/hermenegildosantos) · [github.com/HermenySantos](https://github.com/HermenySantos)

### SUMMARY

Full-Stack AI Engineer with 5+ years shipping production systems across healthcare, enterprise and live events. Primary engineer on Ovee, a live-event AI layer with an operator approving every contribution before it reaches the stage, delivered at 4 live events across Asia and Europe, including a Fortune-500 leadership tour and mci group's CheckedIn 2026, with zero off-message moments on stage. Generalist across backend, frontend, mobile, cloud and AI integration. Stack: TypeScript · React · Python · Go · Azure OpenAI · Node.js.

### TECHNICAL SKILLS

**Languages:** TypeScript, JavaScript, Go, Python, Dart

**Frontend / Mobile:** React, Next.js, React Native (Expo/Reanimated), Flutter, Three.js, Tailwind CSS, Material UI, Redux

**Backend:** Node.js, NestJS, FastAPI, Express, GraphQL, WebSockets, MQTT, WebRTC, RESTful APIs

**AI / LLM:** Azure OpenAI GPT-4o, OpenAI Realtime API, TTS/STT, LLM prompt engineering, human-in-the-loop systems

**Data:** PostgreSQL, MongoDB, Cosmos DB, Firebase, Redis, MySQL — TypeORM, Prisma, SQLAlchemy

**Cloud & DevOps:** Azure (Functions, App Service, OpenAI), AWS (EC2, S3, Lambda), Cloudflare Workers, Docker, GitHub Actions, CI/CD

### EXPERIENCE

#### Full Stack Engineer | Dorier

Jun 2025 – Present

Geneva, Switzerland · Remote

#### Ovee — Live-Event AI (Fortune-500 leadership tour; mci group's CheckedIn 2026)

- Primary engineer (~600 of ~670 commits): operator approval workflow for the moderated stage, FastAPI services, Azure OpenAI GPT-4o generation and TTS voice, React operator and audience apps, WebRTC relay for OpenAI Realtime, WebSocket broadcast
- Delivered at 4 live events across Asia and Europe: a three-summit leadership tour for a Fortune-500 multinational, plus mci group's CheckedIn 2026 in Geneva, where Ovee was billed as a panellist alongside the group CEO; 0 off-message moments on stage
- CheckedIn, 3 event days: ~175,000 requests, 7 server errors, 16–42 ms average response; shipped fixes and a second simultaneous workshop room overnight between sessions
- First summit: an audience spike overwhelmed the S1 plan; scaled live to P2v2 and recovered within ~1 minute, then moved to P2v3 and introduced pre-event load testing
- Workshop synthesis: 60 roundtables / 600 leaders turned into themes on the main display in ~5 s; 699 backend test functions

#### UN Geneva Visitor Centre — Together immersive experience (Fondation Portail des Nations)

- Software engineer on the tour management system, taken over from Local Projects through a nine-week handoff (Jul–Sep 2025) and carried to the public opening in June 2026 (~200,000 visitors/year expected; 3 pavilions, 8 languages, up to 40 visitors per session)
- React Native audio guide, guide tablet and Go tour services over MQTT: in-app Quuppa positioning, audio drift correction (5 s check, 400 ms threshold), headphone reconnection recovery, device-aware tour ownership, group language flows
- Deaf accessibility mode: audio pipeline runs muted so captions stay timed to the show; Gathering proximity effect mapping indoor position (5 m → 0.5 m) to a live visual cue
- Emergency audio end to end (DB migrations, Go service and API, CMS, MQTT, handheld ducking and looping evacuation, offline assets) — built and tested, not yet enabled onsite; wrote most automated tests and the platform's technical documentation

#### Scopely Global Celebration Day — WebAR Character Hunt (sole engineer)

- Only engineer on a browser AR game (no app install) run at 12 Scopely hubs, Tokyo to Mexico, on the same day; designed for ~2,100 players at an event Scopely reported at 2,400 employees

- Built the Mattercraft/Zappar experience, Azure Functions API, Cosmos DB model partitioned by location and the React staff dashboard; five-week core build (Jul 7–Aug 8, 2025) plus three weeks of refinement
- On-device anti-cheat: normalised iOS/Android motion sampling (30/60/100 Hz) with raw-sensor fallback; shake penalised only with walking evidence and  $\geq 200$  ms persistence, tuned for fairness
- Server-side score clamp, 12-location whitelist, profanity filter, 5 submissions/IP/location/hour rate limit, 7-day automatic data deletion (GDPR)

#### Enterprise Automation — Galderma / Lightcraft

- Python CLI/GUI tool with Dropbox integration, Gmail notifications, OAuth2 auth, cross-platform GitHub Actions CI/CD producing standalone executables — in production

#### Project Development Manager / Full Stack Developer | NomadEngenuity

Apr 2024 – Apr 2025

Covilhã, Portugal · Hybrid

- Architected microservice platform (Next.js + NestJS + TypeScript) for Seezy eye care app — improved data processing by 35%, reduced API response times by 40% via query optimisation and caching
- Implemented GraphQL + RESTful APIs handling 1,000+ daily transactions; configured Azure CI/CD with zero-downtime deployments
- Built PharmaSee healthcare staffing front-end (React, WebSockets, advanced matching algorithms); 65% mobile usage, 85% test coverage

#### Software Developer | DC Tech

Mar 2023 – Jun 2023

Covilhã, Portugal · Part-time

- Built award-winning full-stack environmental disaster prevention prototype (React, Node.js, ML models for flood/wildfire prediction); secured incubator funding in Porto

#### Programming Teacher | Conservatório de Música da Covilhã

Oct 2022 – Jun 2023

Covilhã, Portugal · Part-time

- Designed curriculum (Scratch, HTML/CSS, JavaScript) for 100+ students; built browser-based code editor with real-time preview and gamification (85% engagement rate)

#### Software Developer | 2GF Innovation Technology

Jul 2021 – Oct 2022

Remote

- Built React/TypeScript interfaces for cross-border family platform (10,000+ MAU), WebRTC video communication, reusable component library (~40% new-feature time), Flutter/Dart offline-first mobile app, Firebase auth/realtime DB
- Developed Node.js middleware integrating SAP enterprise systems; ReactJS dashboards for operational metrics; MongoDB analytics schemas

### SELECTED PERSONAL PROJECTS

[InvoFlow](#) — FastAPI · Next.js 14 · PostgreSQL · Azure Document Intelligence · Cloudflare R2 · Clerk · Docker

- Full-stack invoice/receipt management SaaS for Portuguese SMBs — OCR-powered data extraction, real-time IVA tracking, accountant-ready export; Docker Compose deployment with mock mode for local dev

[NexTool API](#) — TypeScript · Cloudflare Workers · Hono · Vitest (30 tests) · RapidAPI

- 13+ developer utility endpoints (QR codes, meta scraping, text analytics, hashing, IP geolocation, colour conversion, JSON tools) deployed globally on Cloudflare edge; monetised on RapidAPI marketplace

### EDUCATION

**MBA (ESG Focus)** · University of Beira Interior Executive, Portugal

2024 – Present

**BSc Computer Science** · University of Beira Interior, Portugal · Final project: React + Flutter + Firebase RBAC badge system

2018 – 2021

### LANGUAGES

**Portuguese** — Native | **English** — Full Professional Proficiency (spoken & written)